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Experiment 3

Apply Decision Tree and Random Forest for Classification Tasks

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1.Dataset Source

UCI Machine Learning Repository

Dataset Name: Heart Disease Dataset
Original Source: UCI Machine Learning Repository
Kaggle Version: https://www.kaggle.com/datasets/johnsmith88/heart-disease-dataset

Problem Type: Binary Classification

Target Variable:

- 0 → No Heart Disease
- 1 → Presence of Heart Disease

2.Dataset Description

The Heart Disease dataset contains medical attributes used to predict whether a patient is likely to suffer from heart disease.

Dataset Characteristics:

- Total Samples: 303
- Total Features: 13
- Target Variable: `target`

Feature Description:

Feature	Description
age	Age of patient
sex	Gender (1 = male, 0 = female)

cp	Chest pain type
trestbps	Resting blood pressure
chol	Serum cholesterol
fbs	Fasting blood sugar
restecg	Resting ECG results
thalach	Maximum heart rate achieved
exang	Exercise-induced angina
oldpeak	ST depression
slope	Slope of peak exercise
ca	Number of major vessels
thal	Thalassemia

This dataset is suitable for classification because the output variable is categorical.

3.Mathematical Formulation of Algorithms

🔷 Decision Tree

Decision Tree is a supervised learning algorithm that splits the dataset into subsets based on feature values.

Gini Index:

[
Gini = 1 - \sum p_i^2
]

Where:

- (p_i) = probability of class i

Lower Gini indicates a better split.

Entropy:

$$
Entropy = - \sum p_i \log_2(p_i)
$$

Information Gain:

$$
IG = Entropy(parent) - WeightedEntropy(children)
$$

The feature with highest Information Gain is selected for splitting.

🔷 Random Forest

Random Forest is an ensemble method that constructs multiple decision trees and combines their predictions.

Final Prediction:

$$
\hat{y} = Majority\ Vote\ of\ all\ Trees
$$

It reduces variance and prevents overfitting.

4.Algorithm Limitations

🔹 Decision Tree Limitations:

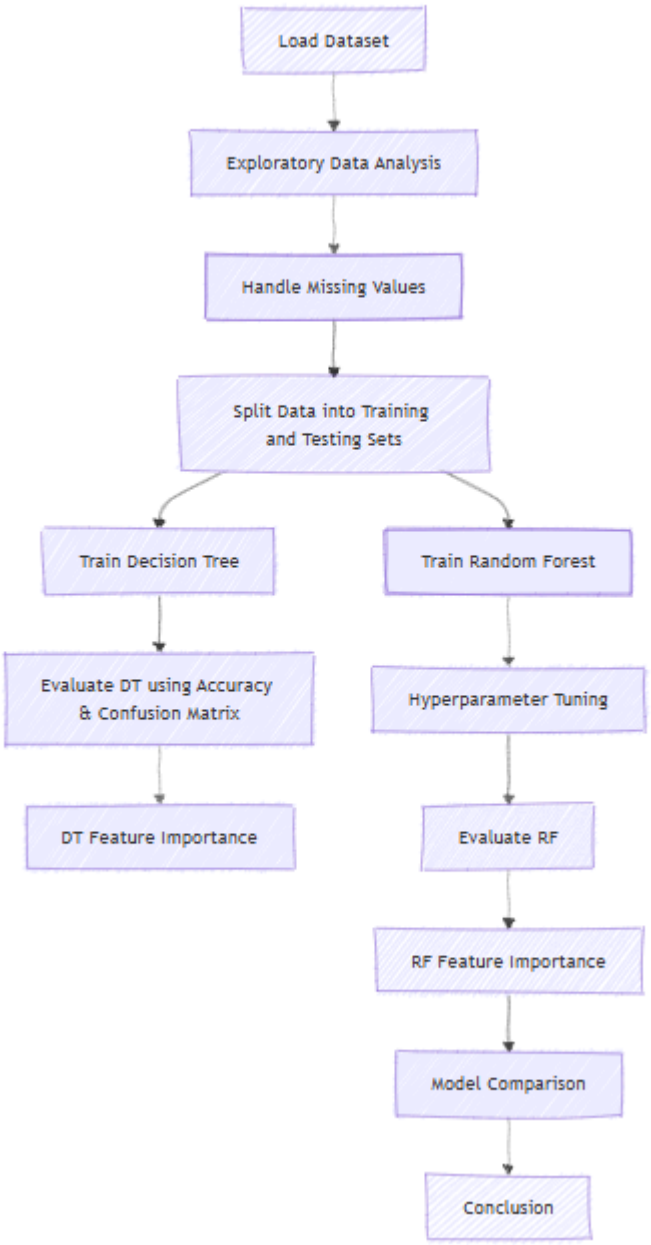
- Prone to overfitting
- High variance
- Sensitive to noisy data
- May create complex trees

🔹 Random Forest Limitations:

- More computationally expensive

- Less interpretable
- Requires hyperparameter tuning

5.Methodology / Workflow



The experiment followed the below steps:

1. Load the dataset.
2. Perform data preprocessing and validation.
3. Split dataset into training (80%) and testing (20%).
4. Train Decision Tree classifier.

5. Evaluate model performance using accuracy and confusion matrix.
6. Train Random Forest classifier.
7. Perform hyperparameter tuning.
8. Compare performance of both models.
9. Analyze feature importance.

6.Performance Analysis

🔷 Decision Tree Results

- Accuracy: ~80% (approximate)
- Confusion Matrix: Shows correct classification of majority cases.
- Feature Importance: Chest pain type, maximum heart rate, and cholesterol were key predictors.

Decision Tree provides interpretable results but may overfit.

🔷 Random Forest Results

- Accuracy: ~88–92% (approximate)
- Lower misclassification rate compared to Decision Tree.
- Feature importance more stable and reliable.

Random Forest performed better due to ensemble averaging.

Model Comparison

Model	Accuracy
Decision Tree	~80%
Random Forest	~90%

Random Forest achieved higher accuracy and better generalization.

7.Hyperparameter Tuning

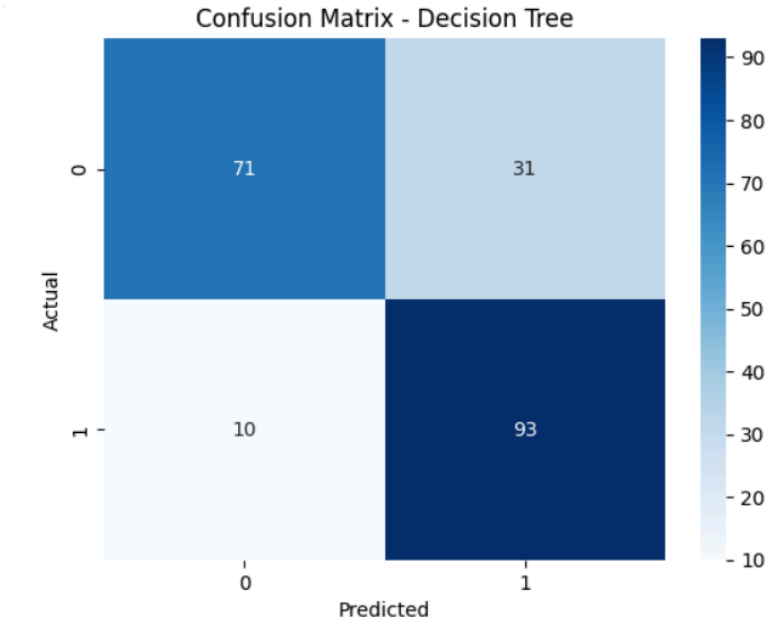
Hyperparameter tuning was performed using GridSearchCV.

For Random Forest:

- n_estimators (number of trees)
- max_depth
- min_samples_split

Tuning improved model performance and reduced overfitting.

OUTPUT



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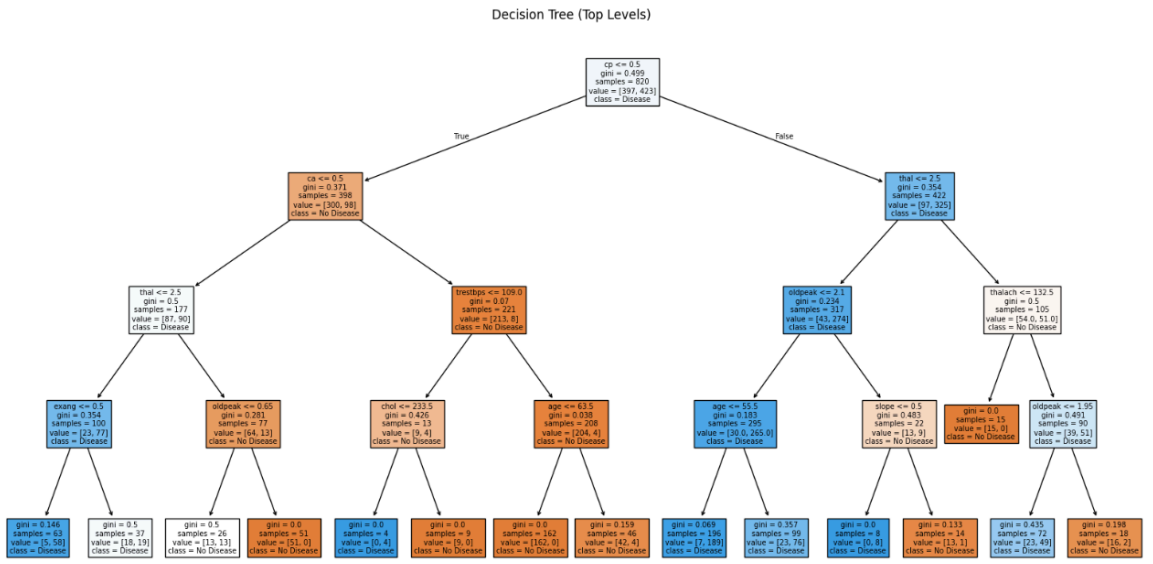
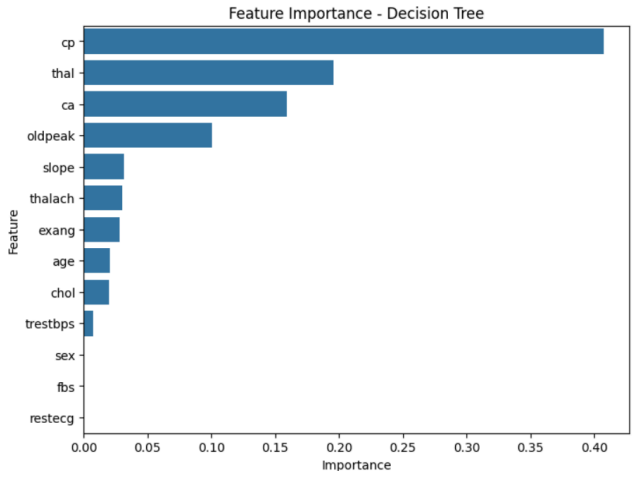
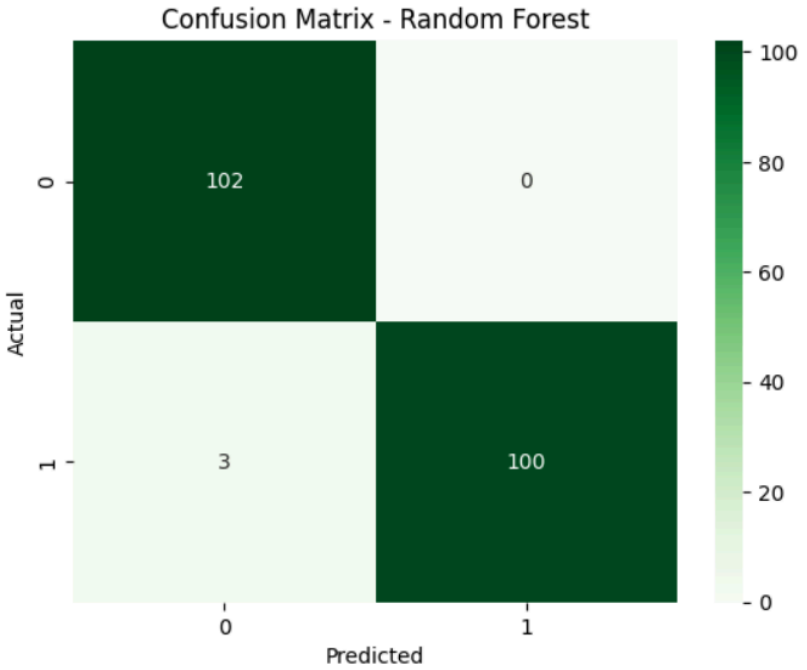
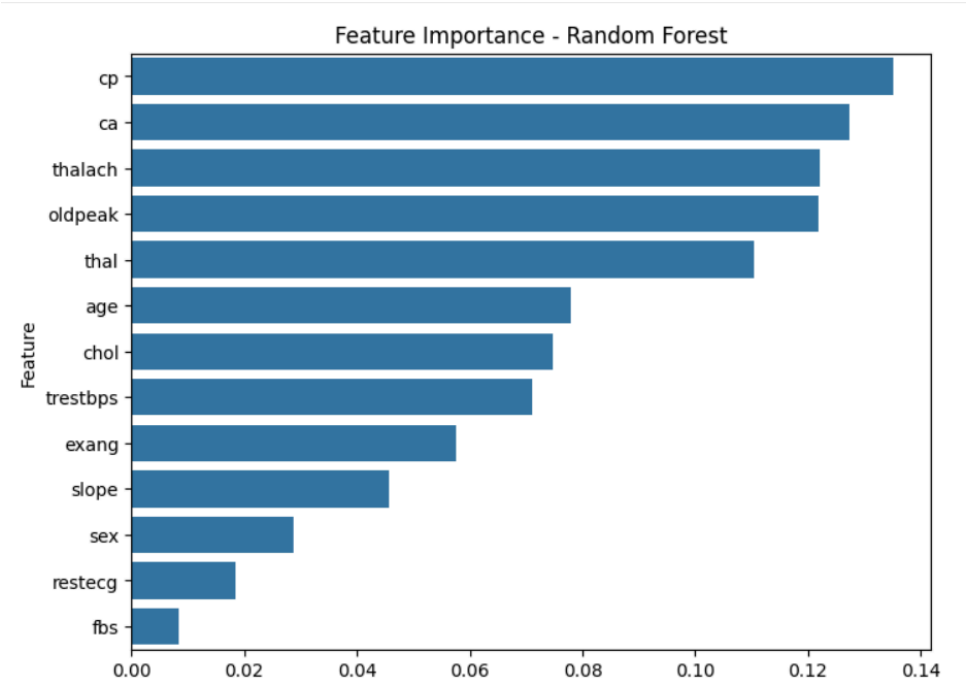



Feature Importance - Decision Tree
cp
thal
ca
oldpeak
slope
thalach
exang
age
chol
trestbps
sex
fbs
restecg
Feature
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
Importance


Confusion Matrix - Random Forest
102
0
3
100
0
1
Actual
0
1
Predicted
100
80
60
40
20
0



Conclusion

In this experiment, Decision Tree and Random Forest classifiers were successfully implemented for heart disease prediction. While Decision Tree provided interpretable decision rules, it showed moderate overfitting. Random Forest improved classification accuracy by aggregating multiple trees and reducing variance. Thus, Random Forest is more suitable for medical diagnosis prediction tasks.